

Post-Event Mechanical Inspection

A synthetic condition report separating gross visual findings from dimensional and functional conclusions.

Document ID	Incident	Record type	Status
R3-EV-013	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Inspection findings

Component	Visual finding	What is not established	Status
Carrier beam	No visible fracture or permanent gross bend in preliminary visual check.	Load history, subtle distortion, and material condition.	Synthetic scenario fact
Link	No visible fracture; one contact edge shows a polished witness area.	When the mark formed and whether it indicates partial seating.	Synthetic scenario fact / interpretation open
Keeper and pin	Keeper present; pin retained.	Keeper engagement before the fall and dimensional condition.	Historical source fact / details open
Yoke pocket	No gross crack or break; contact surfaces require cleaning and measurement.	Wear, taper, clearance, and dynamic path.	Historical source fact / pending
Window liner	Recovered displaced; attachment hardware condition not fully documented.	Whether liner movement preceded or followed beam motion.	Open

Recommended follow-up tests

- Controlled dimensional inspection against the full drawing and tolerance stack.
- Material and non-destructive examination if engineering identifies critical areas.
- Fit-up demonstration using an exemplar or digital model without disturbing original evidence.
- Failure-mode review distinguishing fracture, pin release, link unseating, interference, and support loss.